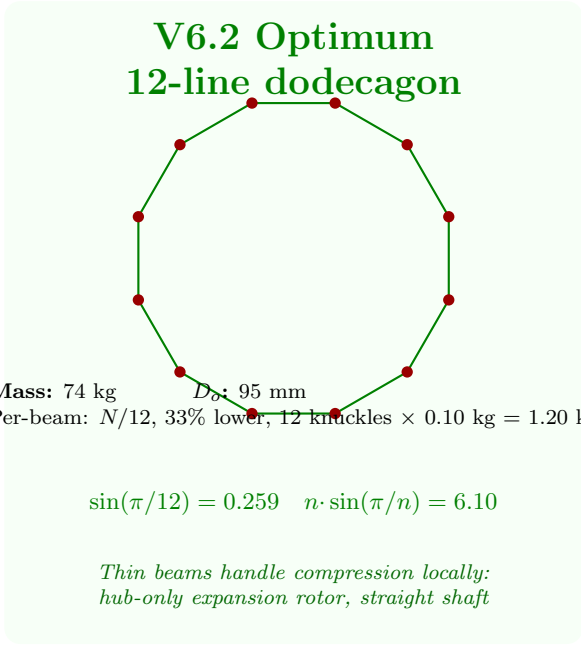
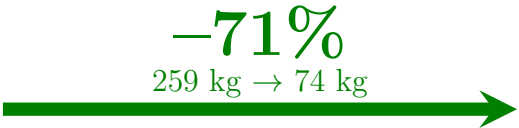
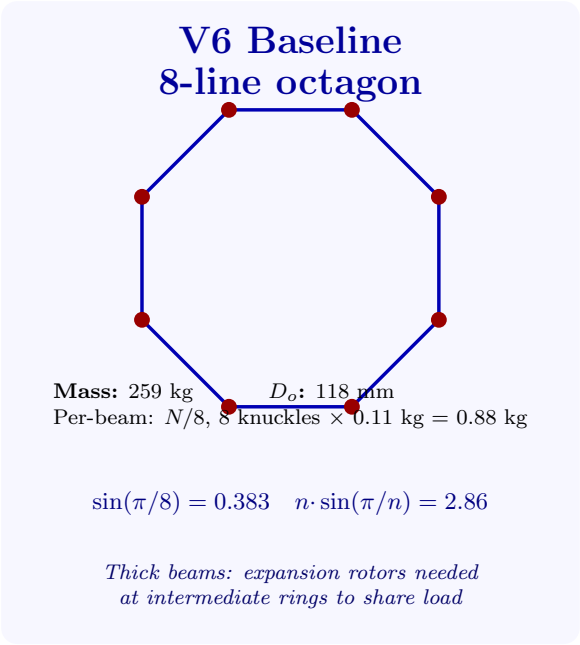


The Polygon Flip: Why 12 Lines Beat 8

Beam-only scaling says fewer lines are lighter. The DE optimiser found the opposite.



What this reveals about the TRPT scaling problem:

The beam-mass-only formula ($n \cdot \sin(\pi/n)$) increases by only 1.20 $\times$ from $n=3$ to $n=12$ – far less than intuition suggests. Thinner beams at higher n enable dramatically smaller knuckle hardware. The DE optimiser evaluates the full coupled design space – beams, knuckles, tethers, rotor sizing, ring count, density profile, expansion stations – and finds the global minimum at $n=12$, not $n=3$ or $n=8$. This is a system-level inversion of a component-level intuition.